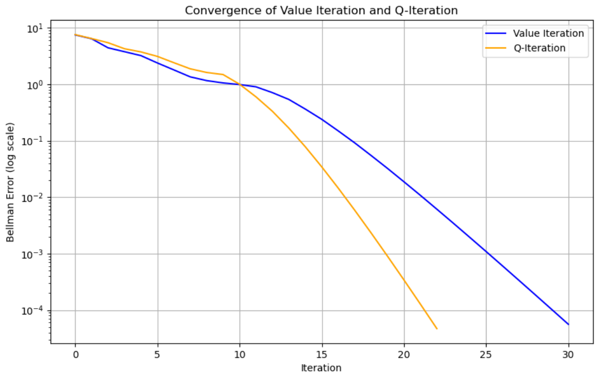
P1:

- Number of iterations until convergence for both algorithms

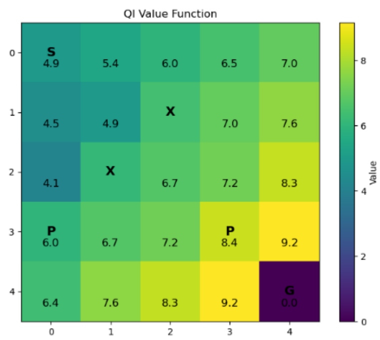


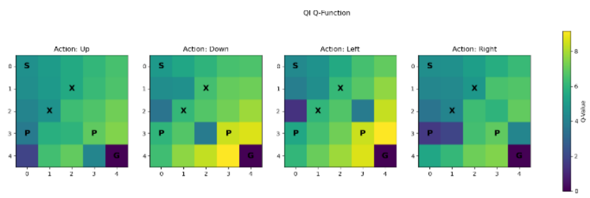
Q-it: 23

Value: 31

- Visualization of the final value function (heatmap)

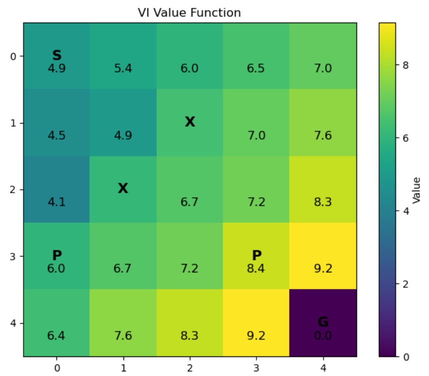
QI Value Function



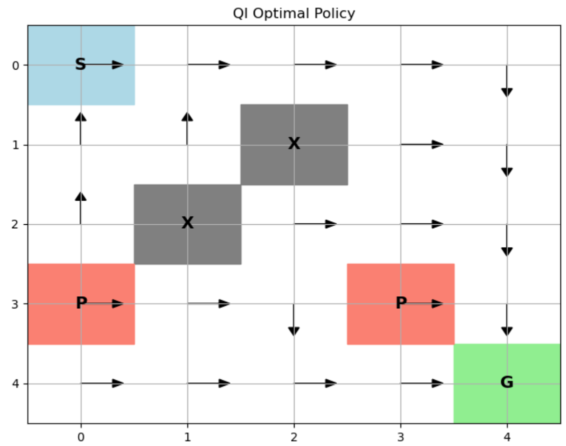


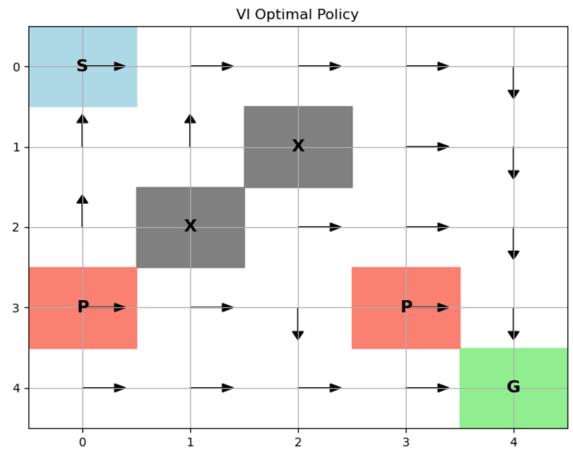
VI value function

•



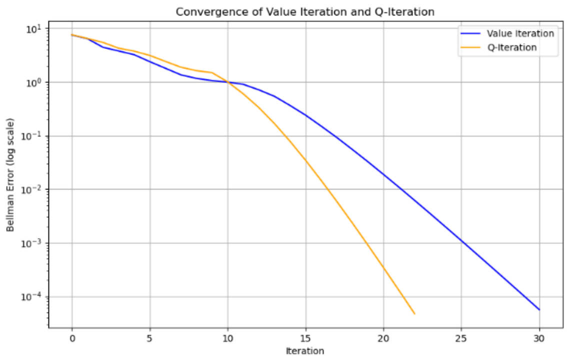
• Visualization of the optimal policy (arrows on grid)





Same

• Brief comparison of Value Iteration vs Q-Iteration convergence rates



At first, they are about the same. After 10 iterations, Q-it converges faster.

- Discussion of how the stochastic transitions affect the optimal policy

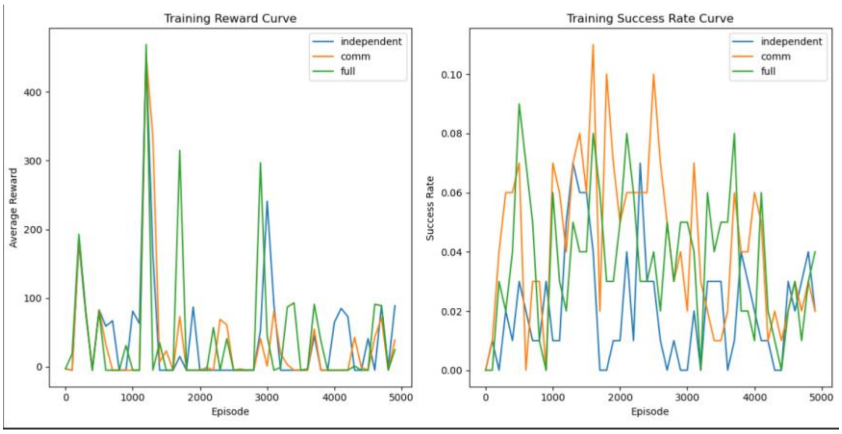
  In deterministic environments, agents seek only the shortest path. However, in stochastic environments, agents seek paths that maximize expected return. This means they sacrifice speed for safety. It prefers open spaces and keep away from the penalty areas.

P2.

- Training hyperparameters (learning rate, batch size, epsilon schedule, replay buffer size)

  Lr = 1e-3, bs = 32, epsilon_start = 1.0, epsilon_end = 0.05, epsilon_decay = 0,995, replay_buffer_size = 10000

- Training curves for all three configurations showing average reward and success rate



- Final success rates for each configuration

  Full: 0.03

  Comm: 0.03

  Independent: 0.01

- Comparison table of performance across configurations
- Analysis of how distance information and communication affect coordination

  Independent: Two agents can only rely on local observations, resulting in poor coordination, low success rates, and longer paths.

  Comm: Agents can exchange signals, partially enhancing collaboration, but lacking global distance perception, coordination remains limited.

  Full: Agents can fully exchange information and perceive each other's distances, achieving optimal collaboration, the highest success rate and rewards, and the shortest path.

- Discussion of learned strategies in each configuration

  Independent: Agents tend to explore independently, struggling to synchronize their arrival at the target, occasionally succeeding randomly.

  Comm: Agents attempt to synchronize actions through communication signals, but coordination remains unstable when distance information is lacking.

  Full: Agents achieve efficient collaboration through communication and distance information, typically actively converging and synchronizing their arrival at the target, demonstrating distinct team strategies.